Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing
Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW – 2 (Model Lab)

Date: 20-04-2026

SMART CAMPUS EVENT MANAGEMENT SYSTEM

SDG: SDG 4 – Quality Education

Presented By:

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INTRODUCTION

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- In many organizations and institutions, event management still depends heavily on manual coordination, phone calls, and repeated message sharing.
- This often leads to missed updates, confusion in scheduling, and poor participant management.
- Smart campus event management system helps all users stay updated without needing repeated manual follow-ups.
- The system allows admin to create and edit events.
- Students can browse a live feed of events and register with just one click.
- Overall, it is a modern platform designed to improve event planning, coordination, and communication.

PROBLEM STATEMENT

- Manual registration processes create high administrative workloads and lead to frequent data entry errors.
- Students lack a centralized digital dashboard to track their personal registration history and schedules.
- There is no real-time analytical system to help administrators monitor student engagement or event popularity.
- Communicating schedule or venue changes manually is slow and often fails to reach all participants in time.
- Manual data management prevents the institution from generating fast and accurate attendance reports.
- Existing methods lack the automation required to ensure a reliable and streamlined event management experience.

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OBJECTIVES

- **To design and develop** a centralized web-based system that automates the entire campus event discovery and registration process.

- **To implement** a real-time analytics dashboard for monitoring student participation and improving overall administrative data accuracy.

PROPOSED SYSTEM

- The proposed system is a centralized web-based platform built with Spring Boot and Thymeleaf for end-to-end campus event management.
- It includes a secure admin dashboard for managing event lifecycles and a dedicated student portal for registrations.
- By automating the registration process, it eliminates the need for manual paperwork and reduces human errors in participant tracking.
- It addresses the lack of insights by providing real-time analytics and data visualizations to monitor student engagement levels.
- The platform offers superior accessibility, allowing users to register for and manage events 24/7 from any device.
- Centralized data storage ensures higher security and integrity of student records compared to manual spreadsheets.
- Automated confirmation and notification systems significantly enhance the speed and reliability of campus-wide communication.

PROPOSED SYSTEM ARCHITECTURE

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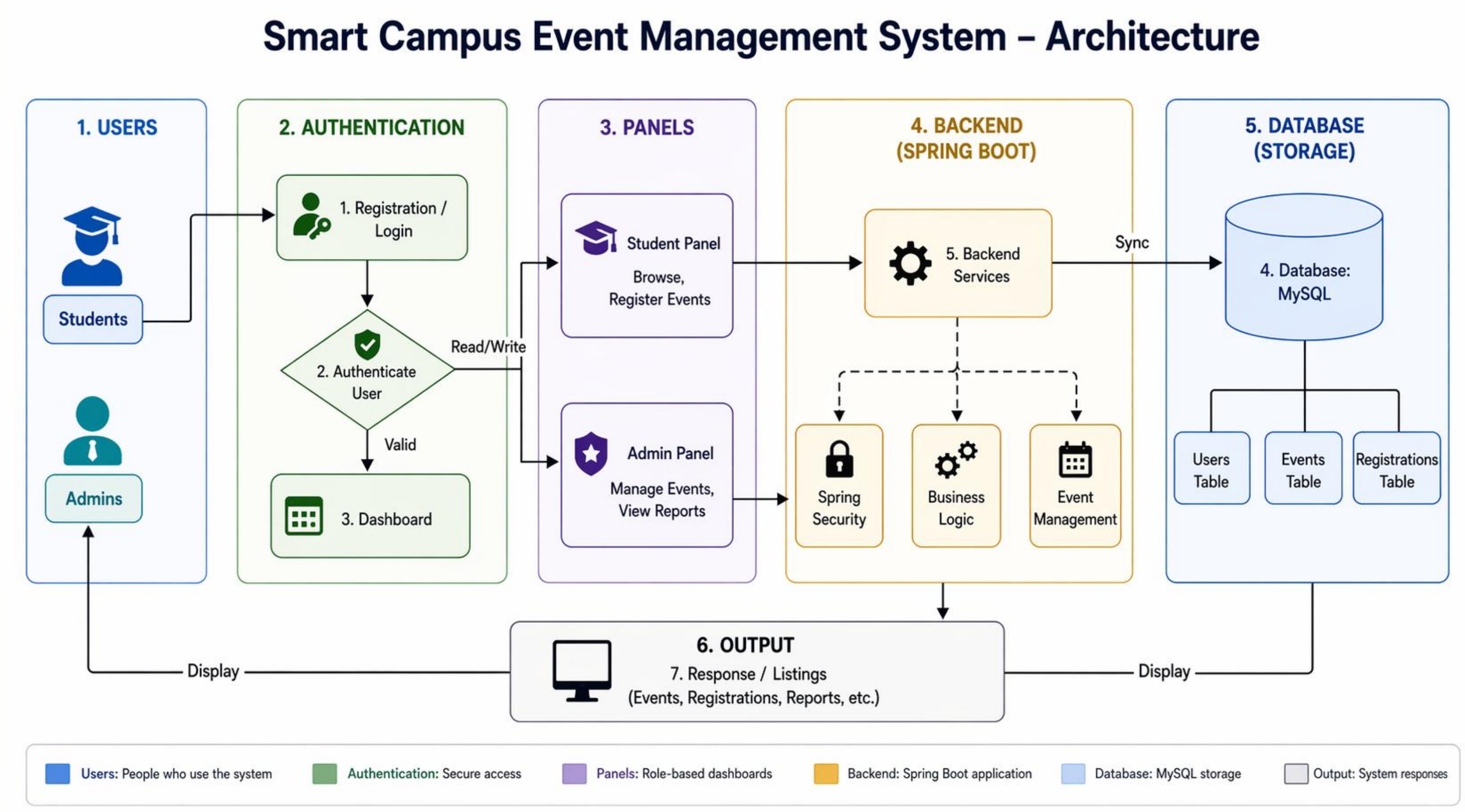

PROPOSED SYSTEM ARCHITECTURE

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- **Input:** User data such as event details, participant invitations, responses, and event updates are entered into the system.
- **Processing:** The system processes the input by handling event creation, updating records, sending notifications, and synchronizing data in real time.
- **Database:** All information related to users, events, participants, and status is stored securely in the database for quick access and management.
- **Output:** The system displays updated event information, participant status, instant notifications, and synchronized results to all users.

WORKFLOW

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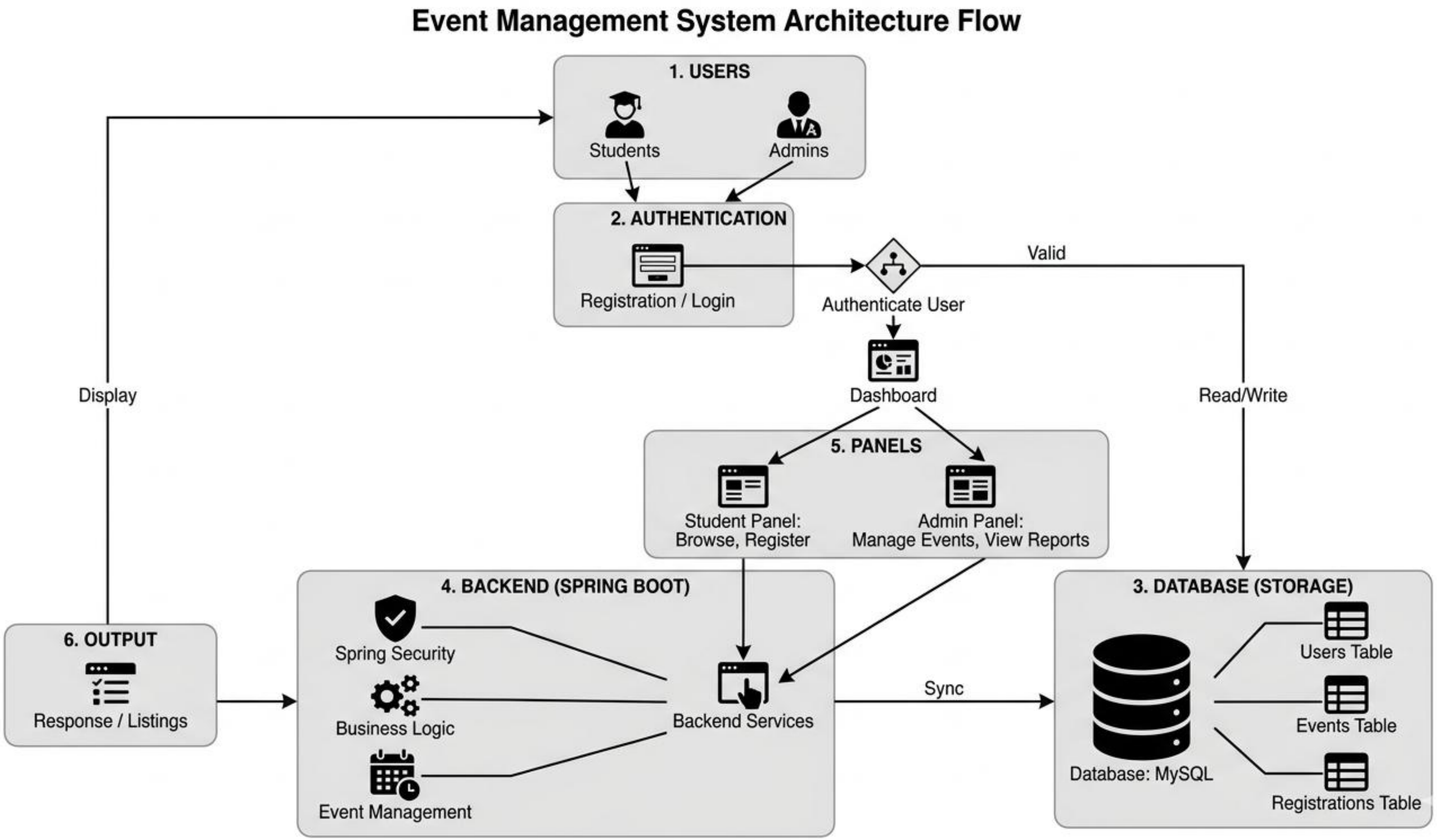

TECHNOLOGIES USED

- **Frontend:** HTML5, CSS3, JavaScript
- **Backend:** Java, Spring Boot
- **Database:** MySQL
- **Tools:** Visual Studio Code, Git, GitHub, Postman, Maven
- **Hardware/Software Requirements:** Computer with minimum 4 GB RAM, dual-core processor, Node.js environment, modern web browser, and operating system such as Windows, Linux, or macOS

MODULE DESCRIPTION

Module 1: User Authentication and Management
Handles student registration, login, secure authentication, and basic profile-related information for accessing the system.

Module 2: Event Management
Admins can create new events, specifying date, time, venue, and descriptions..

Module 3: Administrative Control Hub
Provides administrators with tools to create, edit, or delete events and manage the comprehensive list of participant registrations.

Module 4: Analytics & Visual Reporting
Utilizes Chart.js to generate real-time visual insights into event popularity, registration trends, and student participation metrics.

IMPLEMENTATION SCREENSHOTS

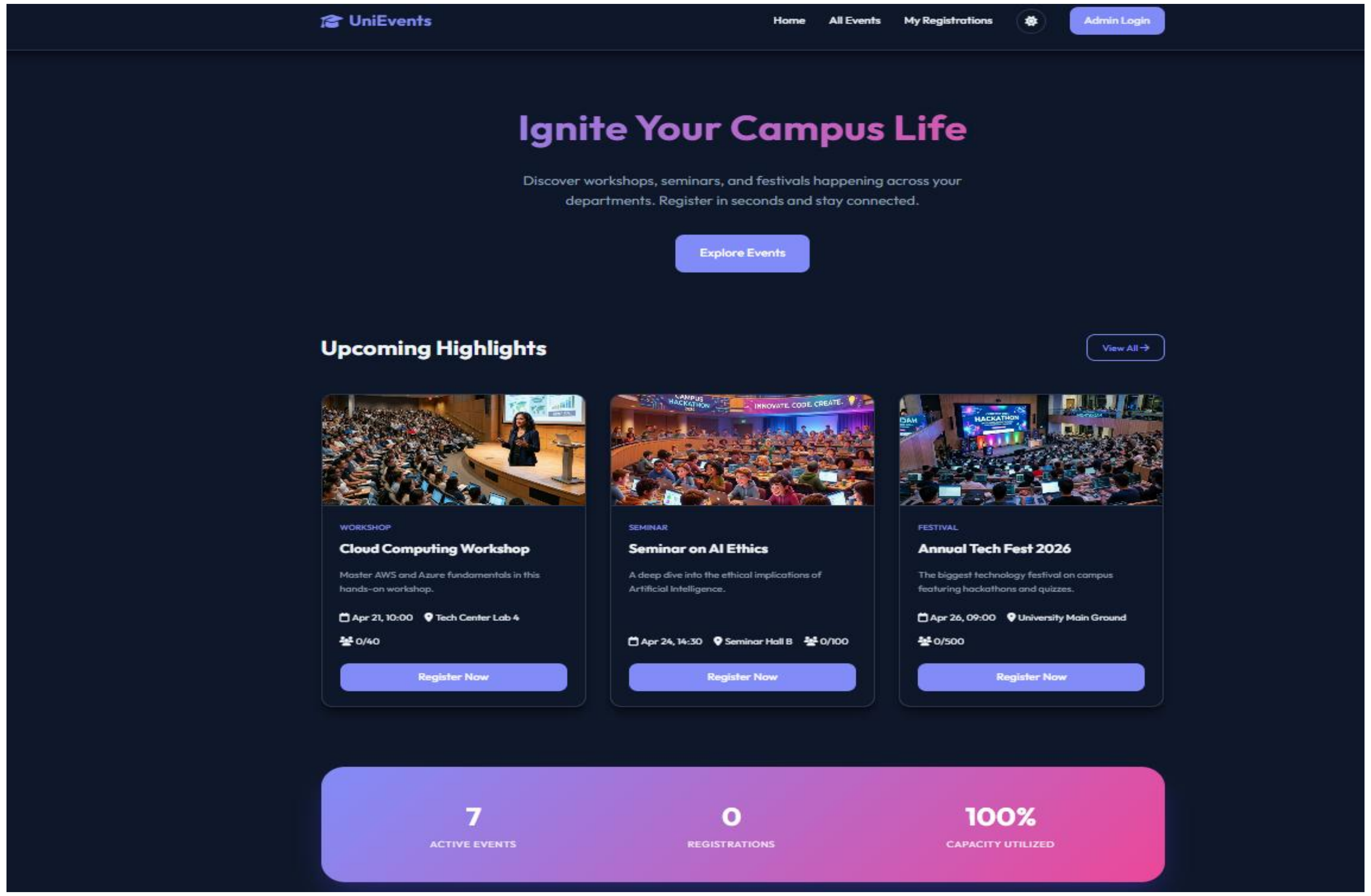

Image 1: Dashboard of the website with details regarding events

IMPLEMENTATION SCREENSHOTS

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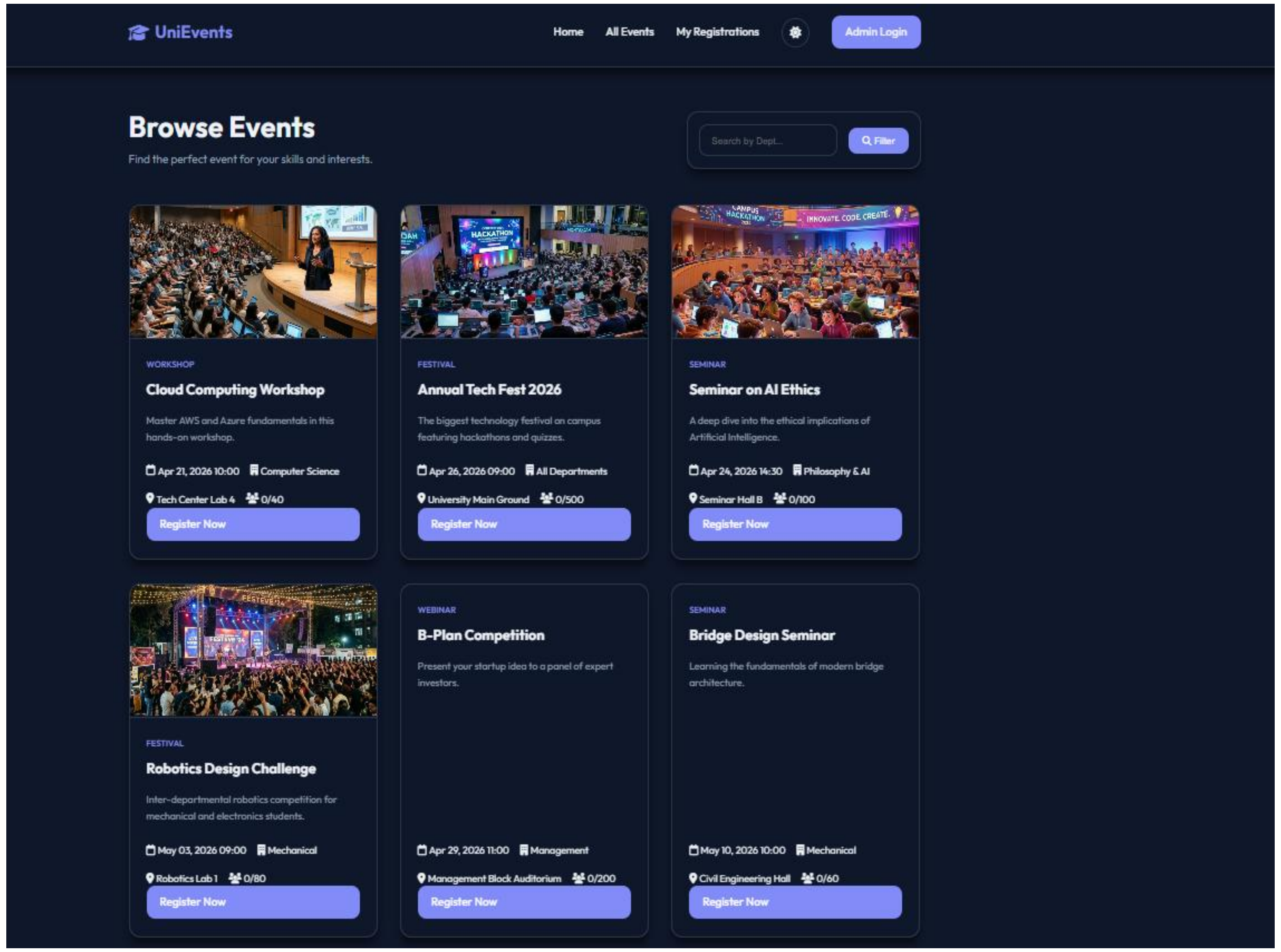

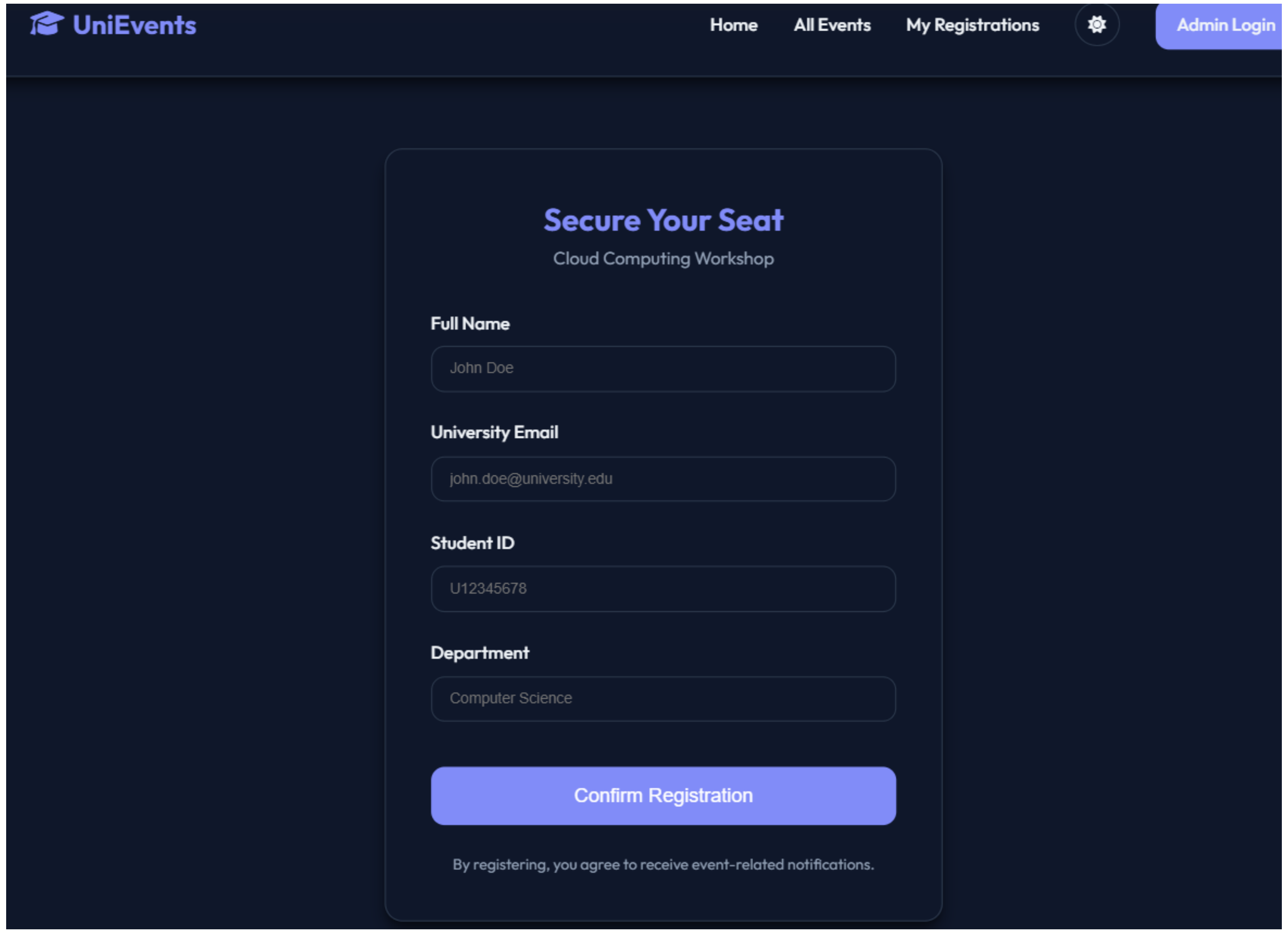

Image 2: Events screen of the student with details regarding available events.

Image 3: Registration screen of student where they can enter credentials.

ADVANTAGES & LIMITATIONS

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Advantages:

- **Centralization**: All campus activities in one unified digital platform.
- **Automation:** Eliminates manual paperwork and administrative errors.
- **Data Insights:** Real-time visual tracking of student engagement.
- **Accessibility:** 24/7 mobile-friendly registration for all students.

Limitations:

- **Connectivity:** Requires a stable internet connection to function.
- **User Adoption**: Effectiveness depends on institutional-wide usage.
- **Maintenance:** Needs regular updates for data security and privacy.
- **Initial Input:** Requires time-consuming initial setup for event data.

Thank you